AQA 3.1.12 Acids and bases

AQA A-level Chemistry · Revision questions · Sheet 1 of 8

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

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20 marks · about 25 minutes

Key information

- $pH = -\log_{10}[H^+]$ and $[H^+] = 10^{-pH}$
- Strong acid: $[H^+]$ = [acid]; for a diprotic strong acid $[H^+] = 2 \times$ [acid]
- $K_w = [H^+][OH^-]$ (1.00×10^{-14} mol^2 dm^{-6} at 25 °C); for a base $[H^+] = K_w \div [OH^-]$
- Pure water is neutral at every temperature because $[H^+] = [OH^-]$; its pH is only 7 at 25 °C
- Weak acid on its own: $[H^+] = \sqrt{(K_a \times [HA])}$; $K_a = 10^{-pKa}$

Section A: Short-answer questions

01 Sulfuric acid is a strong diprotic acid. Calculate the pH of 0.0050 mol dm^{-3} sulfuric acid.

[1 mark]

02 Pure water dissociates slightly: $H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$. The equilibrium constant $K_c = [H^+][OH^-] \div [H_2O]$ and the ionic product of water $K_w = [H^+][OH^-]$

Explain why $[H_2O]$ is not shown in the K_w expression.

[1 mark]

03 Explain why a 0.10 mol dm^{-3} solution of ethanoic acid has a higher pH than a 0.10 mol dm^{-3} solution of hydrochloric acid.

[1 mark]

04 A buffer contains 0.30 mol of benzoic acid and 0.15 mol of sodium benzoate in 1.00 dm^3. Calculate the pH of the buffer. For benzoic acid, $K_a = 6.31 \times 10^{-5}$ mol dm^{-3}

[2 marks]

05 Nitric acid is a strong acid. Calculate the pH of 0.150 mol dm^{-3} nitric acid. Give your answer to 2 decimal places.

[2 marks]

06 Benzoic acid has $K_a = 6.31 \times 10^{-5}$ mol dm^{-3}. Calculate its pK_a.

[1 mark]

Section B: Longer questions

07 Strontium and barium are elements in Group 2.

07.1 Calculate the pH of a 0.0100 mol dm^{-3} solution of strontium hydroxide at 10 °C. You may assume that strontium hydroxide is completely dissociated in this solution.

At 10 °C the ionic product of water, $K_w = 2.93 \times 10^{-15}$ mol^2 dm^{-6}

[3 marks]

07.2 The pH of a barium hydroxide solution is lower at 50 °C than at 10 °C.

At 50 °C a 25 cm^3 sample of this barium hydroxide solution was neutralised by 22.45 cm^3 of hydrochloric acid added from a burette.

Deduce whether the volume of this hydrochloric acid needed to neutralise another 25 cm^3 sample of this barium hydroxide solution at 10 °C is greater than, equal to or less than 22.45 cm^3. Explain your answer.

[2 marks]

08 Hydrochloric acid is a strong acid.

08.1 In an experiment, 10.35 cm^3 of 0.100 mol dm^{-3} hydrochloric acid are added to 25.0 cm^3 of 0.150 mol dm^{-3} barium hydroxide solution.

Calculate the pH of the solution that forms at 30 °C. $K_w = 1.47 \times 10^{-14}$ mol^2 dm^{-6} at 30 °C

Give your answer to 2 decimal places.

[6 marks]

08.2 The pH of water at 30 °C is 6.92. Give the reason why water is neutral at this temperature.

[1 mark]

Mark Scheme

M1, M2... are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

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Question	Marking guidance	Comments	Mark
01	2.00 ($[H^+] = 0.010$ mol dm^{-3})	*Source: Practice question (written for this set)*	1
02	$[H_2O]$ is (very) large in comparison (to $[H^+]$ and $[OH^-]$) / $[H_2O]$ is (almost) constant / incorporated in K_w	*Source: 7405/1 Nov 2021 Q06.1* Allow $K_w = K_c[H_2O]$ Ignore $[H_2O] = 1$ or very small	1
03	Ethanoic acid only partially dissociates so $[H^+]$ is lower	*Source: Practice question (written for this set)*	1
04	M1 $[H^+] = 6.31 \times 10^{-5} \times 0.30 \div 0.15 = 1.26 \times 10^{-4}$ M2 pH = 3.90	*Source: Practice question (written for this set)*	2
05	M1 pH = $-\log_{10}[H^+] = -\log_{10}(0.150)$ M2 pH = 0.82	*Source: 7405/1 Jun 2025 Q08.1* Correct value to 2 dp scores 2 marks	2
06	4.20	*Source: Practice question (written for this set)*	1
07.1	M1 $[OH^-] = 2 \times 0.0100 = 0.0200$ M2 $[H^+] = 2.93 \times 10^{-15} \div 0.0200 = 1.47 \times 10^{-13}$ M3 pH = 12.83	*Source: 7405/1 Jun 2022 Q04.2–04.3* Allow 12.8 If pH = 12.53 allow 2 marks (factor of 2 not used)	3
07.2	M1 = 22.45 cm^3 M2 same $[OH^-]$ / same amount of OH^- ions		2
08.1	M1 $n(HCl) = 1.035 \times 10^{-3}$ **and** $n(Ba(OH)_2) = 3.75 \times 10^{-3}$ M2 $n(OH^-) = 2 \times 3.75 \times 10^{-3} = 7.50 \times 10^{-3}$ M3 excess $OH^- = 7.50 \times 10^{-3} - 1.035 \times 10^{-3} = 6.465 \times 10^{-3}$ mol in 35.35 cm^3 M4 $[OH^-] = 6.465 \times 10^{-3} \div 35.35 \times 10^{-3} = 0.1829$ mol dm^{-3} M5 $[H^+] = K_w \div [OH^-] = 1.47 \times 10^{-14} \div 0.1829 = 8.04 \times 10^{-14}$ M6 pH = 13.09 to 13.10	*Source: 7405/1 Jun 2018 Q05.2–05.3* If M1 incorrect lose M1 and M6; if not × 2 lose M2 and M6; no subtraction lose M3 and M6 If $K_w = 1.00 \times 10^{-14}$ used, lose M5 only (ECF pH = 13.26) Must be 2 dp	6
08.2	$[H^+] = [OH^-]$		1